

Chapter 1 checkpoint!

Hello and welcome to your first checkpoint! Here come five questions, one about each of the learning targets from Chapter 1. This is your scorecard:

Learning target:	1.3	1.5	1.4-A	1.4-G	1.6	1.8
Your confidence level before starting (0-5):						
Your confidence level after the quiz (0-5):						
The mark you earned on this attempt:	Success! Try again!	Success! Try again!	Success! Try again!	Success! Try again!	Success! Try again!	Success! Try again!

Before anything else, please do the following:

- Rank your confidence from 0-5 on each of the learning targets. 5 means “I could teach a whole class about this;” 0 means “I am genuinely not sure I have heard these words before.”
- Write your name on this page and on each of the other pages of the quiz.

Then do the quiz! Some reminders:

- Open notes, closed computer.
- If you need more room to write, use the back of the same learning target page, or ask me for some scratch paper.
- Read the questions carefully and make sure you’re answering each part.
- Show all your work and explain all your thinking!

When you are done:

- Rank your confidence from 0-5 on each of the learning targets. 5 means “I absolutely nailed that question for sure;” 0 means “oof, I definitely didn’t get that one.”
- Make double sure your name is on every page, including any scratch paper.
- Hand in your work, separated by learning target.

Have fun and do your best! I believe in u ♥

Learning targets 1.3 and 1.5

Arapaho Glacier is a mountain glacier in Roosevelt National Forest, west of Boulder, CO. The following table¹ gives the surface area, $A(t)$, in square meters, of Arapaho Glacier in the year t .

t	1900	1960	1973	1999
$A(t)$	338,282	250,764	225,000	162,027

- (a) Compute an approximation for $A'(1960)$, and **include units** for this number.
- (b) Write a sentence explaining what the number from part (a) means about the glacier.
Don't say "per," and don't say "rate."
- (c) Compute an approximation for $A'(1999)$, and **include units** for this number.
Do you think your approximation is too high or too low? Why?
- (d) (Bonus: answer on back) How does $A'(1960)$ compare to $A'(1999)$? Is that good or bad?

¹Haugen, B., Scambos, T., Pfeffer, T., & Anderson, R. (2010). Twentieth-century changes in the thickness and extent of Arapaho Glacier, Front Range, Colorado. *Arctic, Antarctic, and Alpine Research*, 42(2), 198-209.

Learning target 1.4-A

Use the limit definition of the derivative to find a derivative function algebraically.

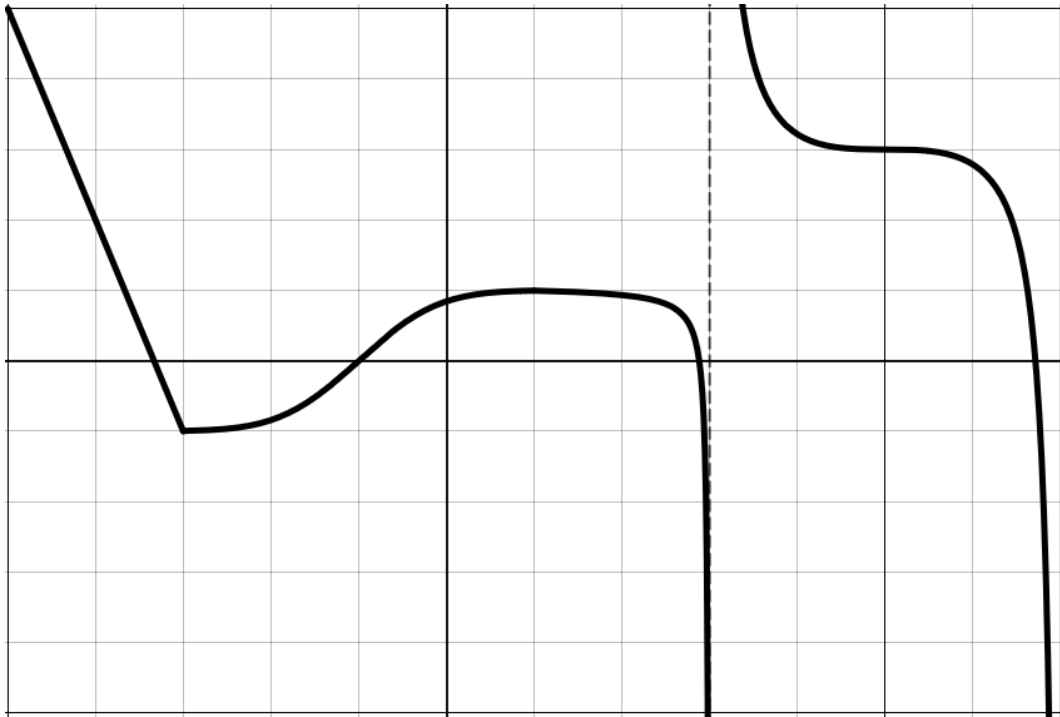
Suppose that $f(x) = 3x^2 - 5x + 4$. Use the limit definition of the derivative to find $f'(x)$.

PS: The answer is $f'(x) = 6x - 5$.

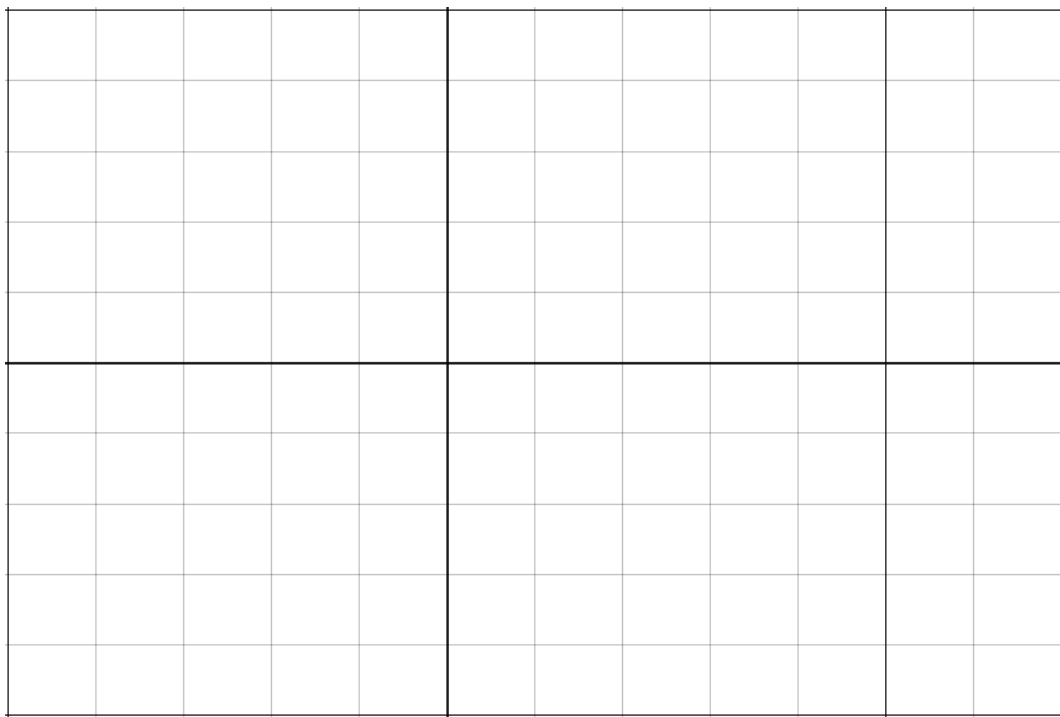
Learning target 1.4-G

Given a graph of a function, estimate specific values of the derivative, and/or produce a sketch of the derivative function.

Here is the graph of some wacky function $q(x)$:



Sketch the graph of $q'(x)$ on the blank axes below.



Learning target 1.6

Explain the relationships between a function, its derivative, and its second derivative. Interpret information about these objects in real-world contexts.

Students in a recent calculus class were playing a Mad Libs game about the relationships between shapes of functions and values of the derivative:

Whenever the _____ is _____, the _____ is _____.
 noun adjective noun adjective

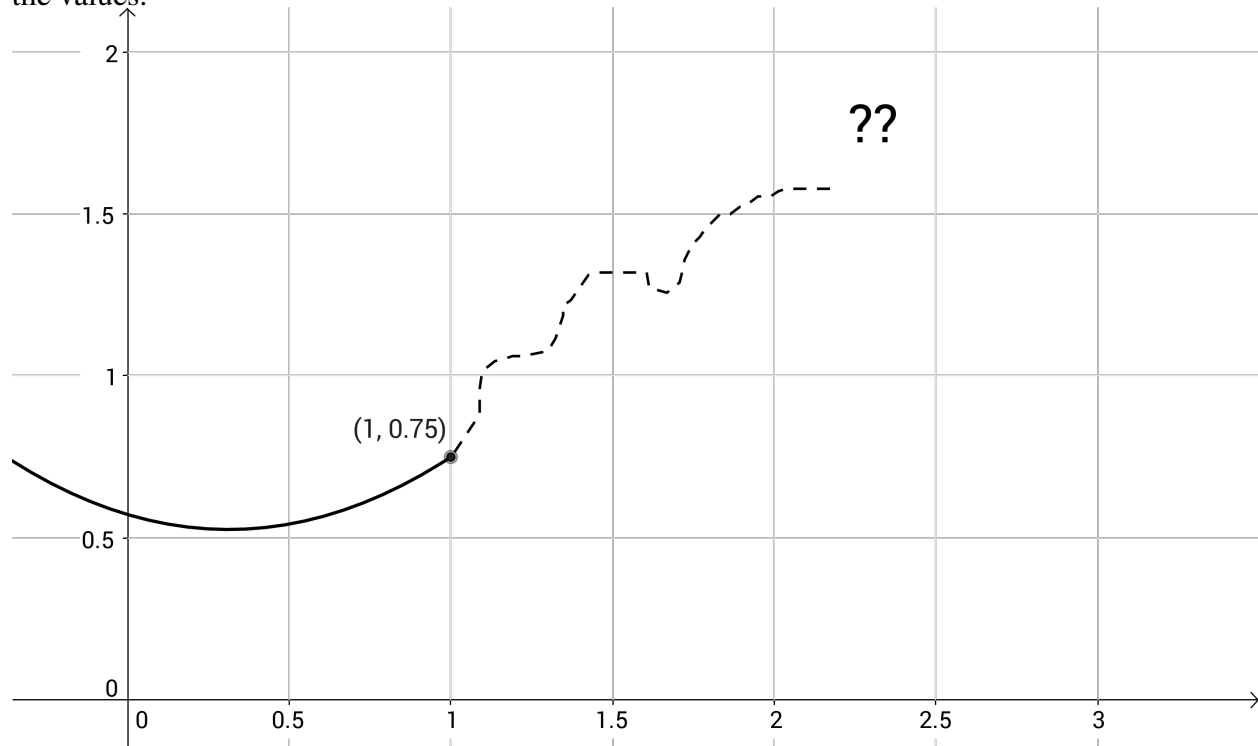
For each of the sentences they generated below, explain why it is correct or incorrect.
(Hint: I don't think it's possible to generate a good explanation without saying the word "slope.")

- (a) Whenever the original function is decreasing, the derivative is negative.
- (b) Whenever the derivative is increasing, the original function is positive.
- (c) Whenever the second derivative is positive, the original function is concave up.
- (d) Whenever the derivative is increasing, the second derivative is positive.

Learning target 1.8

Find the equation for the tangent line to a function at a given point, and use tangent lines to approximate function values.

Part of the graph of this function $f(x)$ has been erased. We'll use linear approximation to recover the values.



- (a) Draw the tangent line to this function at $x = 1$.
- (b) Suppose we know that $f'(1) = 0.65$. Use point-slope form to write down an equation for $L(x)$, the tangent line to this function at $x = 1$.
- (c) Use your equation for $L(x)$ to estimate $f(1.4)$.
- (d) Do you think it is more likely that this is an overestimate or an underestimate? How come?